QP CODE: 24900026

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FIRST SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATION NOVEMBER 2024
First Semester
Discipline Specific Core Course - MGIDSCBIF100 - FUNDAMENTAL IT FOR BIOINFORMATICS
(2024 ADMISSION ONWARDS)

Duration: 2 Hours Maximum Marks: 50

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I),
Appreciation (Ap), and Skill (S)

Students should attend at least one question from each course 0utcome to enhance their overall outcome attainability.

[Learning Domain][CO No(s)]

Part A
Short Answer Questions
Answer any 5 questions.
Each question carries 2 marks

1. State the key features of the fourth generation of computers. [K] [1]
   - Fourth generation computers were faster, more accurate, and more reliable.
   - They were smaller, more portable, and required less electricity.
2. State the function of a cache memory.[K] [1]
   - It stores frequently accessed data and instructions.
   - It is the fastest memory in a computer, allowing for faster data retrieval.
3. Describe the role of a DNS server.[U] [2]
   - A DNS server's primary role is to translate domain names into IP addresses
   - Which allow browsers to get to websites and other internet resources.
4. List two advantages of using a high-level programming language.[U] [3].
   - High-level languages are easier to read, write, debug, and maintain than low-level languages.
   - They can be compiled or translated to run on many different computer architectures.
   - Enables faster development and prototyping.
5. Name the primary purpose of XML. [K] [3]
   - It is to store and share data across systems in a way that is both human- and machine-readable.
   - XML uses a set of rules to encode documents
   - Predefined rules make it easy to transmit data as XML files over any network
6. Name two advantages of using CSS in web development.[K] [3]
   - CSS allows for user-friendly formatting and makes web pages more visually appealing.
   - It allows for global changes to be made through a single style sheet.
7. Identify difference between genomics and proteomics.[K] [4]
   - Genomics studies the genome.
   - Complete set of genes in a cell or organism.
   - While proteomics studies the proteome.
   - Complete set of proteins produced by a cell or organism.

Part B

Short Essay Type Questions
Answer any 4 questions.
Each question carries 5 marks

8. Compare and contrast the characteristics and functions of RAM and ROM. [U] [1]
   - Defining RAM and ROM (2 marks)
   - The key differences might include Volatility, Modifiability, purpose, speed, Cost, Capacity. RAM is volatile memory and ROM is non-volatile memory. The speed of ROM is slower when compared to RAM. (3 marks)
9. Outline the differences between XML and HTML [K] [3]
   - XML is designed to store and transport data
   - HTML is designed to display data and define the structure of web pages.
   - XML allows users to create custom tags to suit specific data needs.
   - HTML has a predefined set of tags designed for web content. (5 marks)
10. Explain how the *if statement* works in JavaScript with an example. [U] [3]
    - The condition in IF statement evaluates to true, the code block inside the if statement is executed.
    - If the condition evaluates to false, the code block is skipped.(2marks)
    - Syntax- (2 marks)
    - Example- (1 mark)
11. Define text formatting in CSS and explain its significance. [K] [3]
    - CSS properties can be used to decorate text.(2marks)
    - Text Formatting Properties in CSS must include: Font-family, font-size, font-style, font-weight. Text-align, line-height, letter-spacing, word-spacing, Colour, text-shadow. (3marks)
12. Explain the role of Bioinformatics in medicine and agriculture. [UI] [4]
    - Role of Bioinformatics in Medicine: Drug discovery and development , Personalized medicine, genomic studies, disease diagnosis,(2.5 marks)
    - Role of Bioinformatics in Agriculture: Crop development, genetic engineering, food security. (2.5 marks)

13 Discuss the role of biological database in bioinformatics [U] [4]
    - A biological database is a collection of biological data, such as genetic and protein sequences, annotations etc. (1.5 mark)
    - Key Roles of Biological Databases: Data Storage and Organization, Data Sharing and Accessibility, Sequence Analysis, Annotation and Interpretation, Evolutionary Studies. (3.5 marks)

Part C

Long Essay Type Questions.
Answer any 2 questions
Each question carries 10 marks

14 Discuss the different layers of the OSI model. [K] [2]
    - OSI Model is a conceptual framework that standardizes network communication by dividing it into seven layers.(3 marks)
    - Explaining levels including Physical Layer, Data-Link Layer, Network Layer, Transport Layer, Session Layer, Presentation Layer, and Application Layer. (7marks)

15 Explain the structure of an HTML table and describe the use of tags. [U] [3]
    - An HTML table is used to organize and display data in a tabular format consisting of rows and columns. (3marks)
    - Table Tag: <table>, Row Tag: <tr>, Header Cell Tag :<th>, Data Cell Tag: <td>. (5marks)

- Example (2marks)

16. Explain how bioinformatics evolved from early 90's and what the major milestones in its development. [U] [4]

- Emergence of Genomic Data(1990s)
- Development of Biological Databases (1990)
- Focus on Sequence Analysis(1990)
- Completion of the Human Genome Project (2003)
- Launch of Specialized Databases(2000s) (5marks)

Milestones and key advancements in Bioinformatics

- GenBank and Biological Databases
- The advent of next-generation sequencing (NGS)
- Prediction of Protein Structure (5marks)

17 Explain human genome project, its primary goals and achievements. [U] [4].

- Human genome project was an international scientific research project which got successfully completed in the year 2003. (4 marks)
- Goals of the human genome project: Sequencing the entire genome. Identification of the complete human genome. (3marks)
- Achievements: Entire genome is made up of 3164.7 million base pairs. On average, a gene is made up of 3000 nucleotides, other features of human genome. (3marks).

END OF THE QUESTION PAPER